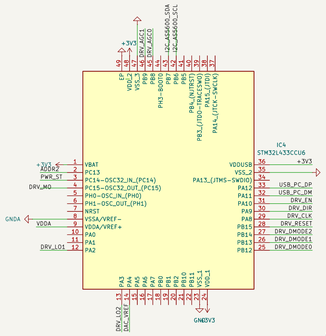
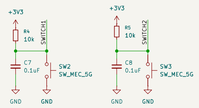
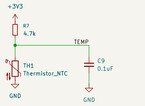










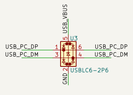




AS5600 external sensor / 3.3V I2C
STM32 I2C1: PB6=SCL, PB7=SDA (AF4)
J5: 1=3V3, 2=GND, 3=SCL, 4=SDA
Fit one set of pull-ups on the bus.



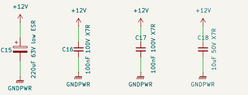








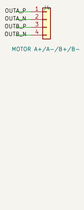
USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9: GPIO VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_CR2.USV in firmware.
Internal 24 pull-up / termination. 90-ohm differential PCB pair.



VM capacitors: C15 bulk 220uF/63V; C16 at VMA; C17 at VMB; C18 10uF/50V.
Keep VM–PGND loops short; paired pins joined at IC on PCB.



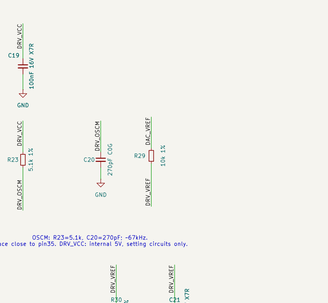






NT1: single PGND–GND copper connection directly beneath IC2 exposed pad.
EP/corner pads to GND; dense thermal vias; 4 layers / 2oz. Motor outputs: 5A routing.

Firmware: DAC1_OUT1 PA4 buffer enabled; reference VDDA/VREF+ (+3V3_A).
Buffered DAC may retain ~0.2V. ENABLE Low or disable DAC for zero current.
Startup: check VM -> set DAC -> set DMODE -> wait >=7.5us -> ENABLE High.
Reset: ENABLE/DMODE/AGC Low; evaluate initially with AGC0=AGC1=Low.


IC2
TB6752249FTG,EL
DRV_AGC0
DRV_AGC1
DRV_CLIM0
DRV_CLIM1
DRV_FLIM
DRV_BOOST
DRV_LTH
DRV_OSCM
DRV_VCC
DRV_VREF
GND
+12V